



Practice Set – Probability

Union of Events, Conditional Probability,
Multiplication Rule, Bayes' Theorem

STA201 – Theory of Probability and Statistics

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Section A: Questions 1–20

1. A batch of 200 machine parts has 50 with surface defects (A), 40 with dimensional defects (B), and 15 with both defects. A part is selected at random. Find
 - (a) $P(A \cup B)$
 - (b) $P(A' \cap B')$
 - (c) $P(A | B)$
 - (d) Are A and B independent?
2. A survey of 120 engineering students found that 70 use Python, 55 use MATLAB, and 30 use both. A student is selected at random. Find the probability that the student
 - (a) uses at least one of the two software tools
 - (b) uses neither
 - (c) uses Python given that the student uses MATLAB
 - (d) uses MATLAB given that the student does not use Python.
3. A system has three independent components in series with reliabilities 0.9, 0.8, and 0.7.
 - (a) What is the probability that the system works?
 - (b) What is the probability that at least one component fails?
4. A diagnostic test has sensitivity $P(+ | D) = 0.95$ and specificity $P(- | D') = 0.90$. The prevalence of the disease is $P(D) = 0.02$. If a person tests positive, what is the probability that they actually have the disease?
5. Two fair dice are rolled. Let A : sum is 8, B : first die is even. Find $P(A)$, $P(B)$, $P(A \cap B)$, $P(A \cup B)$, and $P(A | B)$. Are A and B independent?
6. A card is drawn from a standard deck. Let E : face card, F : heart. Find $P(E)$, $P(F)$, $P(E \cap F)$, $P(E \cup F)$, and $P(E | F)$. Are E and F independent?
7. A lot contains 30 items, of which 6 are defective. Three items are drawn at random without replacement. What is the probability that all three are non-defective?
8. In a company, 60% of employees are engineers and 40% are non-engineers. Among engineers, 20% are managers; among non-engineers, 30% are managers. A randomly selected employee is a manager. What is the probability that the employee is an engineer?
9. A factory has machines M_1 , M_2 , and M_3 , which produce 25%, 35%, and 40% of items. Their defective rates are 1%, 2%, and 3%. If a randomly selected item is defective, what is the probability that it was produced by M_2 ?
10. A biased coin has $P(H) = 0.6$. It is tossed three times. Find the probability that

- (a) all three tosses are heads (b) all three tosses are tails (c) at least one head occurs.
11. On a rainy day, the probability of a traffic accident is 0.08; on a non-rainy day, it is 0.03. The probability of rain on any day is 0.25. If an accident occurs, what is the probability that it rained?
12. A student is late if the bus is late (probability 0.4) or if the student oversleeps (probability 0.3). These events are independent.
- (a) What is the probability that the student is late? (b) What is the probability that the student is not late?
13. A box contains 5 red, 4 blue, and 3 green balls. Two balls are drawn with replacement. Find the probability that
- (a) both are red blue same colour.
 (b) one is red and one is (c) both balls are the
14. In a health survey of 200 people, 80 have high blood pressure (H), 60 have high cholesterol (C), and 30 have both. Find
- (a) $P(H \cup C)$ (d) $P(C | H)$
 (b) $P(H' \cap C')$ (e) Are H and C independent?
 (c) $P(H | C)$
15. A company receives chips from suppliers A , B , and C in proportions 45%, 35%, and 20%. The defective rates are 2%, 3%, and 1%. A randomly selected chip is defective. What is the probability that it came from supplier B ?
16. In a multiple-choice test, a student knows the answer with probability 0.7. If the student knows, the answer is always correct; if not, the answer is guessed correctly with probability 0.25. Given that the student answered correctly, what is the probability that the student actually knew the answer?
17. A circuit has two independent parallel branches. Branch 1 works with probability 0.9, and branch 2 works with probability 0.8. The circuit works if at least one branch works. Find the reliability of the circuit.
18. A jar contains 10 white and 6 black balls. Two balls are drawn without replacement. Find the probability that
- (a) both balls are black black first was white.
 (b) the first ball is white (c) the second ball is
 and the second is black given that the

19. In a software project, the probability of a delay due to requirements changes is 0.2, due to testing issues is 0.15, and due to both is 0.05.
- (a) What is the probability that the project is delayed due to at least one reason? (c) Given a testing delay, what is the probability that there is also a requirements delay?
- (b) What is the probability that there is no delay due to either reason? (d) Are the two delay events independent?
20. An email filter: 95% of spam emails contain the word “offer”, while only 5% of legitimate emails contain this word. Overall, 20% of all emails are spam. If an email contains “offer”, what is the probability that it is spam?

Section B: Questions 21–39

21. A statistics class consists of 25 industrial, 10 mechanical, 10 electrical, and 8 civil engineering students. A student is randomly selected. Find the probability that the student is
- (a) an industrial engineering major (b) a civil or electrical engineering major.
22. In a poker hand of 5 cards, find the probability of holding 2 aces and 3 jacks.
23. What is the probability of getting a total of 7 or 11 when a pair of fair dice is tossed?
24. A box contains 500 envelopes: 75 contain \$100, 150 contain \$25, and 275 contain \$10. An envelope may be purchased for \$25.
- (a) Give the sample space for the different amounts of money.
 (b) Assign probabilities to the sample points.
 (c) Find the probability that the first envelope purchased contains less than \$100.
25. In a senior class of 500 students, 210 smoke, 258 drink alcoholic beverages, 216 eat between meals, 122 smoke and drink, 83 eat between meals and drink, 97 smoke and eat between meals, and 52 engage in all three. A student is selected at random. Find the probability that the student
- (a) smokes but does not drink alcoholic beverages (b) eats between meals and drinks alcoholic beverages but does not smoke (c) neither smokes nor eats between meals.
26. A fuse box contains 20 fuses, of which 5 are defective. Two fuses are selected at random without replacement. What is the probability that both are defective?
27. One bag contains 4 white and 3 black balls; a second bag contains 3 white and 5 black balls. One ball is drawn from the first bag and placed unseen in the second bag. What is the probability that a ball now drawn from the second bag is black?

28. Let R be the event that a convict committed armed robbery, and D be the event that the convict pushed dope. State in words what probabilities are expressed by
- (a) $P(R \mid D)$ (b) $P(D' \mid R)$ (c) $P(R' \mid D')$.
29. For married couples in a certain suburb, the probability that the husband votes is 0.21, the probability that the wife votes is 0.28, and the probability that both vote is 0.15. Find the probability that
- (a) at least one member of the couple votes that the husband votes given that the wife does not vote.
 (b) the wife votes, given (c) the husband votes,
30. A flu vaccine manufacturer processes serum batches through three departments with rejection rates 0.10, 0.08, and 0.12 sequentially and independently. Find
- (a) the probability that a batch survives the first inspection but is rejected by the second (b) the probability that a batch is rejected by the third department.
31. The probability that a doctor correctly diagnoses a particular illness is 0.7. Given that the doctor makes an incorrect diagnosis, the probability that the patient files a lawsuit is 0.9. What is the probability that the doctor makes an incorrect diagnosis and the patient sues?
32. A town has two fire engines operating independently. The probability that a specific engine is available when needed is 0.96.
- (a) What is the probability that neither is available? (b) What is the probability that a fire engine is available?
33. In an assembly plant, machines B_1 , B_2 , and B_3 produce 30%, 45%, and 25% of products. Their defective rates are 2%, 3%, and 2%. A finished product is randomly selected. What is the probability that it is defective?
34. A firm uses plans P_1 , P_2 , P_3 for 30%, 20%, and 50% of products. Defect rates: $P(D|P_1) = 0.01$, $P(D|P_2) = 0.03$, $P(D|P_3) = 0.02$. A randomly selected product is defective. Which plan is most likely responsible?
35. A paint-store chain sells latex and semigloss paint. The probability a customer buys latex paint is 0.75. Of those who buy latex, 60% also buy rollers; only 30% of semigloss buyers purchase rollers. A randomly selected buyer purchases a roller and a can of paint. What is the probability that the paint is latex?
36. An allergist claims that 50% of her patients are allergic to some weed. What is the probability that

- (a) exactly 3 of her next 4 patients are allergic to weeds? (b) none of her next 4 patients is allergic to weeds?
37. The probability that a person makes a mistake on a tax return is 0.1. Find the probability that
- (a) four unrelated persons each make a mistake, and Mr. Roberts and Ms. Williams do not.
(b) Mr. Jones and Ms. Clark both make
38. A manufacturer studies cooking temperature (3 levels), cooking time (4 levels), and type of oil (3 levels) for making potato chips.
- (a) What is the total number of combinations to be studied?
(b) How many combinations will be used for each type of oil?
(c) Discuss why permutations are not an issue.
39. From a box containing 6 black and 4 green balls, 3 balls are drawn with replacement. Find the probability that
- (a) all 3 are the same colour (b) each colour is represented.

Solutions to Non-Immediate Questions

Question 4 (Bayes): $P(+) = 0.95 \times 0.02 + 0.10 \times 0.98 = 0.117$

$$P(D | +) = \frac{0.019}{0.117} \approx 0.1624.$$

Question 8 (Bayes): $P(\text{manager}) = 0.6 \times 0.2 + 0.4 \times 0.3 = 0.24$

$$P(\text{engineer} | \text{manager}) = \frac{0.12}{0.24} = 0.5.$$

Question 9 (Bayes): $P(D) = 0.25 \times 0.01 + 0.35 \times 0.02 + 0.40 \times 0.03 = 0.0215$

$$P(M_2 | D) = \frac{0.007}{0.0215} \approx 0.3256.$$

Question 11 (Bayes): $P(\text{accident}) = 0.08 \times 0.25 + 0.03 \times 0.75 = 0.0425$

$$P(\text{rain} | \text{accident}) = \frac{0.02}{0.0425} \approx 0.4706.$$

Question 15 (Bayes): $P(D) = 0.45 \times 0.02 + 0.35 \times 0.03 + 0.20 \times 0.01 = 0.0215$

$$P(B | D) = \frac{0.0105}{0.0215} \approx 0.4884.$$

Question 16 (Bayes): $P(\text{correct}) = 0.7 \times 1 + 0.3 \times 0.25 = 0.775$

$$P(\text{knows} | \text{correct}) = \frac{0.7}{0.775} \approx 0.9032.$$

Question 20 (Bayes): $P(\text{offer}) = 0.2 \times 0.95 + 0.8 \times 0.05 = 0.23$

$$P(\text{spam} | \text{offer}) = \frac{0.19}{0.23} \approx 0.8261.$$

Question 22 (Poker): Total hands = $\binom{52}{5} = 2,598,960$.

$$\text{Favourable} = \binom{4}{2} \binom{4}{3} = 24.$$

$$\text{Probability} = 24/2,598,960 = 1/108,290 \approx 0.000009234.$$

Question 27 (Transfer & draw): $P(B) = \frac{4}{7} \cdot \frac{5}{9} + \frac{3}{7} \cdot \frac{6}{9} = \frac{38}{63} \approx 0.6032.$

Question 29(c): $P(H \cap W') = 0.21 - 0.15 = 0.06$, $P(W') = 0.72$.

$$P(H | W') = 0.06/0.72 = 1/12 \approx 0.0833.$$

Question 30(b): Batch must survive first two: $0.90 \times 0.92 = 0.828$; then rejected by third: $0.828 \times 0.12 = 0.09936$.

Question 34 (Bayes with 3 plans): $P(D) = 0.30 \times 0.01 + 0.20 \times 0.03 + 0.50 \times 0.02 = 0.019$.

$$\text{Posterior: } P(P_1 | D) = 0.003/0.019 = 0.1579,$$

$$P(P_2 | D) = 0.006/0.019 = 0.3158,$$

$$P(P_3 | D) = 0.010/0.019 = 0.5263.$$

$\Rightarrow$ Plan 3 is most likely.

Question 35 (Bayes): $P(\text{roller}) = 0.75 \times 0.60 + 0.25 \times 0.30 = 0.525$

$$P(\text{latex} \mid \text{roller}) = \frac{0.45}{0.525} = \frac{6}{7} \approx 0.8571.$$

Question 36 (Binomial): (a) $\binom{4}{3}(0.5)^3(0.5)^1 = 4 \times 0.125 \times 0.5 = 0.25$

(b) $(0.5)^4 = 0.0625.$

Question 38: (a) $3 \times 4 \times 3 = 36$ combinations.

(b) $3 \times 4 = 12$ per oil.

(c) Permutations are not relevant because the order of factor levels does not change the recipe.

Final Answer Key

Section A

1. (a) 0.375 (b) 0.625 (c) 0.375 (d) not independent
2. (a) 0.7917 (b) 0.2083 (c) 0.5455 (d) 0.5
3. (a) 0.504 (b) 0.496
4. 0.1624
5. $P(A) = 5/36$, $P(B) = 1/2$, $P(A \cap B) = 1/12$, $P(A \cup B) = 5/9$, $P(A|B) = 1/6$; not independent
6. $P(E) = 3/13$, $P(F) = 1/4$, $P(E \cap F) = 3/52$, $P(E \cup F) = 11/26$, $P(E|F) = 3/13$; independent
7. ≈ 0.4985
8. 0.5
9. ≈ 0.3256
10. (a) 0.216 (b) 0.064 (c) 0.936
11. ≈ 0.4706
12. (a) 0.58 (b) 0.42
13. (a) $25/144$ (b) $5/18$ (c) $25/72$
14. (a) 0.55 (b) 0.45 (c) 0.5 (d) 0.375 (e) not independent
15. ≈ 0.4884
16. ≈ 0.9032
17. 0.98
18. (a) $1/8$ (b) $1/4$ (c) $2/5$
19. (a) 0.30 (b) 0.70 (c) $1/3$ (d) not independent
20. ≈ 0.8261

Section B

1. (a) $25/53 \approx 0.4717$ (b) $18/53 \approx 0.3396$
2. $1/108,290 \approx 0.000009234$
3. $2/9 \approx 0.2222$
4. (a) $\{\$10, \$25, \$100\}$ (b) $P(\$10) = 0.55$, $P(\$25) = 0.30$, $P(\$100) = 0.15$ (c) 0.85

5. (a) 0.176 (b) 0.062 (c) 0.342
6. $1/19 \approx 0.0526$
7. $38/63 \approx 0.6032$
8. (a) Prob. armed robbery given pushed dope (b) Prob. did not push dope given armed robbery (c) Prob. no armed robbery given did not push dope
9. (a) 0.34 (b) $5/7 \approx 0.7143$ (c) $1/12 \approx 0.0833$
10. (a) 0.072 (b) 0.09936
11. 0.27
12. (a) 0.0016 (b) 0.9984
13. 0.0245
14. Plan 3; posterior: $P_1 : 0.1579$, $P_2 : 0.3158$, $P_3 : 0.5263$
15. $6/7 \approx 0.8571$
16. (a) 0.25 (b) 0.0625
17. (a) 0.0001 (b) 0.0081
18. (a) 36 (b) 12 (c) permutations not relevant
19. (a) $7/25 = 0.28$ (b) $18/25 = 0.72$